RATAN TATA MAHARASHTRA STATE SKILLS UNIVERSITY

School of Science Engineering and Technology Skills

Department of Engineering

Subject – AWS Lab

Experiment – 5

Title: Secure VPC Infrastructure Setup

Lab Objective:
To design and implement a Virtual Private Cloud (VPC) with public and private subnets, configure automated routing for internet access via NAT and Internet Gateways, and apply multi-layered security using Security Groups and NACLs.

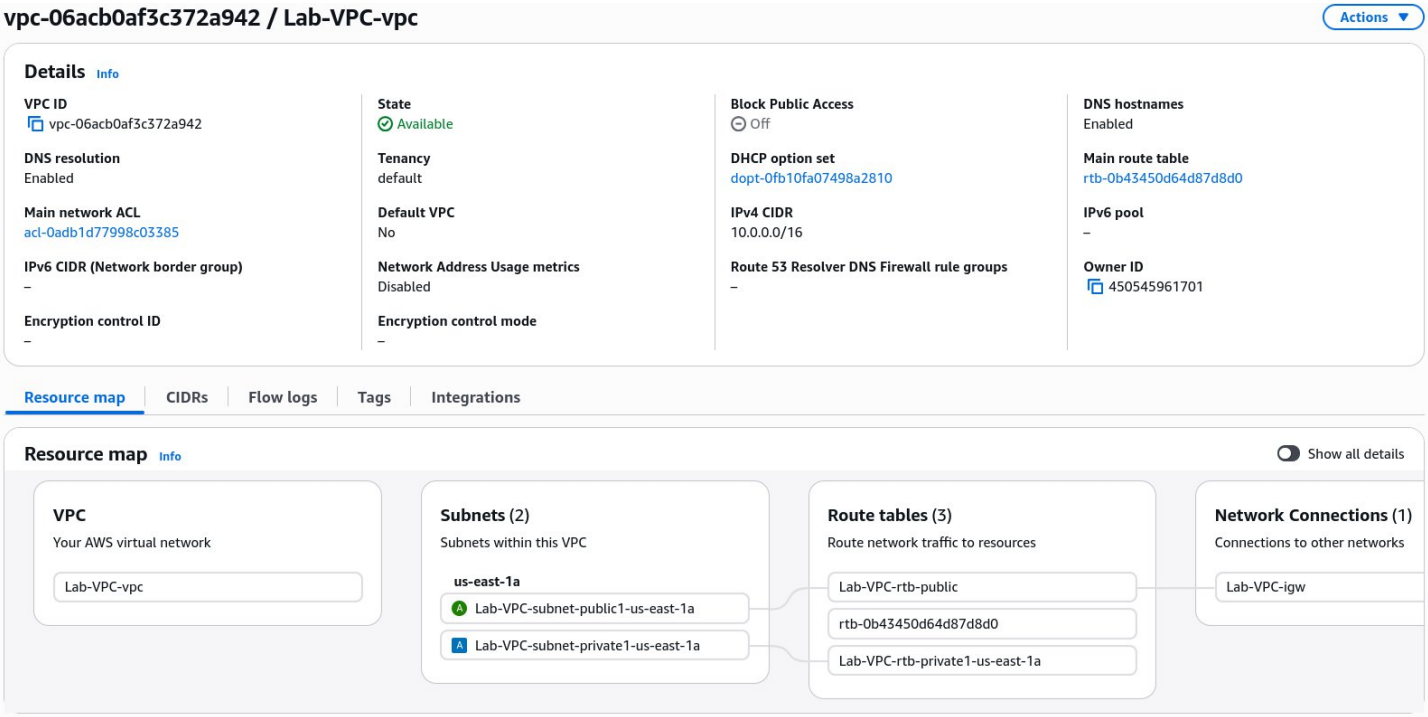
Prerequisites:
- An active AWS Account.
- Basic understanding of CIDR notation (e.g., 10.0.0.0/16).

Step 1: Create VPC using the VPC Wizard

The modern "VPC and more" wizard automates the creation of subnets, route tables, and gateways in one workflow.

1. Log in to the AWS Management Console and navigate to VPC.
2. Click Create VPC.
3. Select "VPC and more" (this is the 2026 equivalent of the VPC Wizard).
4. Configure Name & CIDR:
    - Name tag auto-generation: Lab-VPC
    - IPv4 CIDR block: 10.0.0.0/16
5. Configure Subnets:
    - Number of Availability Zones (AZs): 1 (for lab simplicity) or 2 (for high availability).
    - Number of Public Subnets: 1
    - Number of Private Subnets: 1

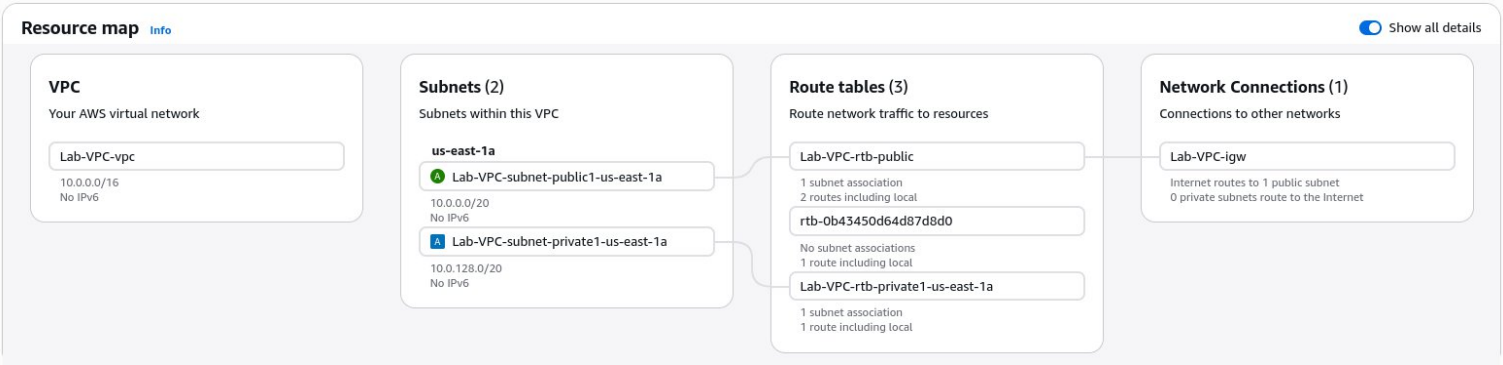
6. Configure Gateways:
    - NAT Gateway: Select 1 per AZ. (Note: This incurs costs; select "None" if just testing routing logic without actual internet requirements).
    - VPC Endpoints: S3 Gateway (Optional, keep as None for this lab).
    - DNS Options: Ensure "Enable DNS hostnames" and "Enable DNS resolution" are checked.
7. Click Create VPC. Wait for the "Success" status across all resources.



Step 2: Component Verification

Once the wizard finishes, verify the resources created:

- Subnets: Check for one subnet with Public in the name and one with Private.
- Internet Gateway (IGW): Ensure it is attached to your Lab-VPC.
- Route Tables:
    - Public Route Table: Should have a route 0.0.0.0/0 → igw-xxxx.
    - Private Route Table: Should have a route 0.0.0.0/0 → nat-xxxx.

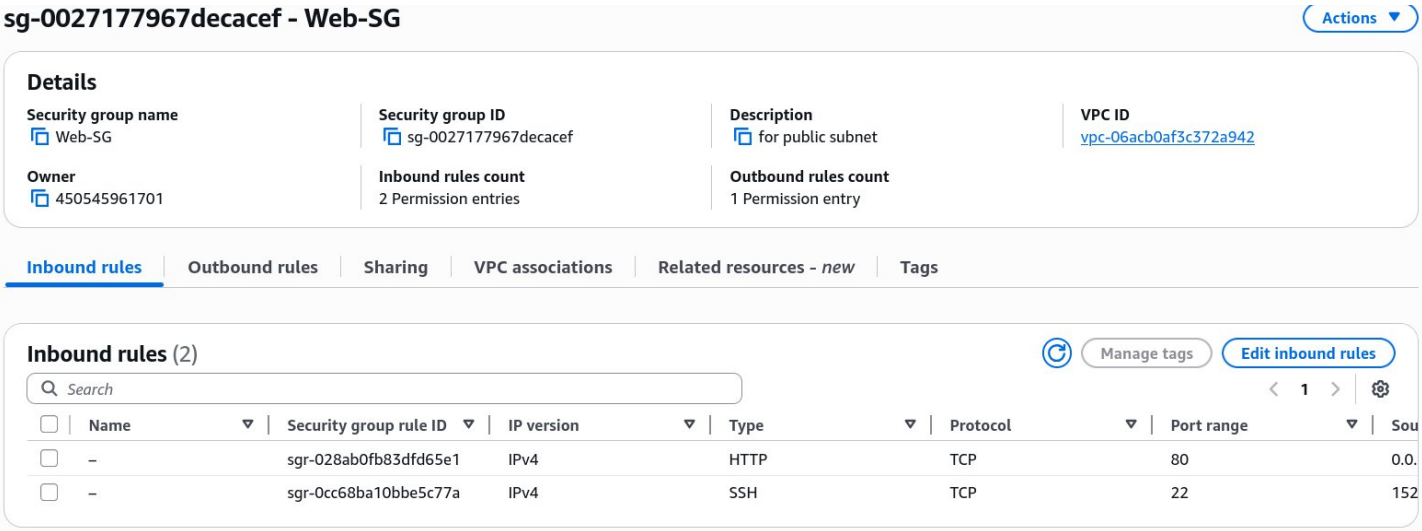


Step 3: Configure Network Security

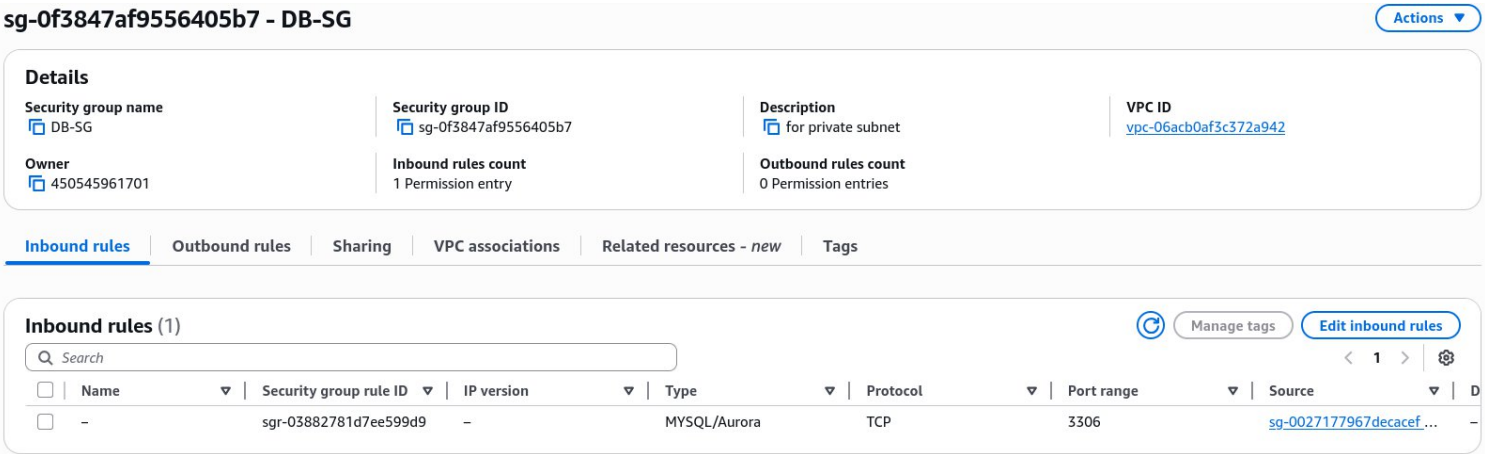
A. Security Groups (Instance-Level Security)

Security Groups are stateful; if you allow inbound traffic, outbound response is automatically allowed.

1. In the VPC Dashboard, go to Security Groups > Create security group.
2. Web-SG (for Public Subnet):
   - Inbound Rules:
     - Type: HTTP (80) | Source: 0.0.0.0/0
     - Type: SSH (22) | Source: Your IP



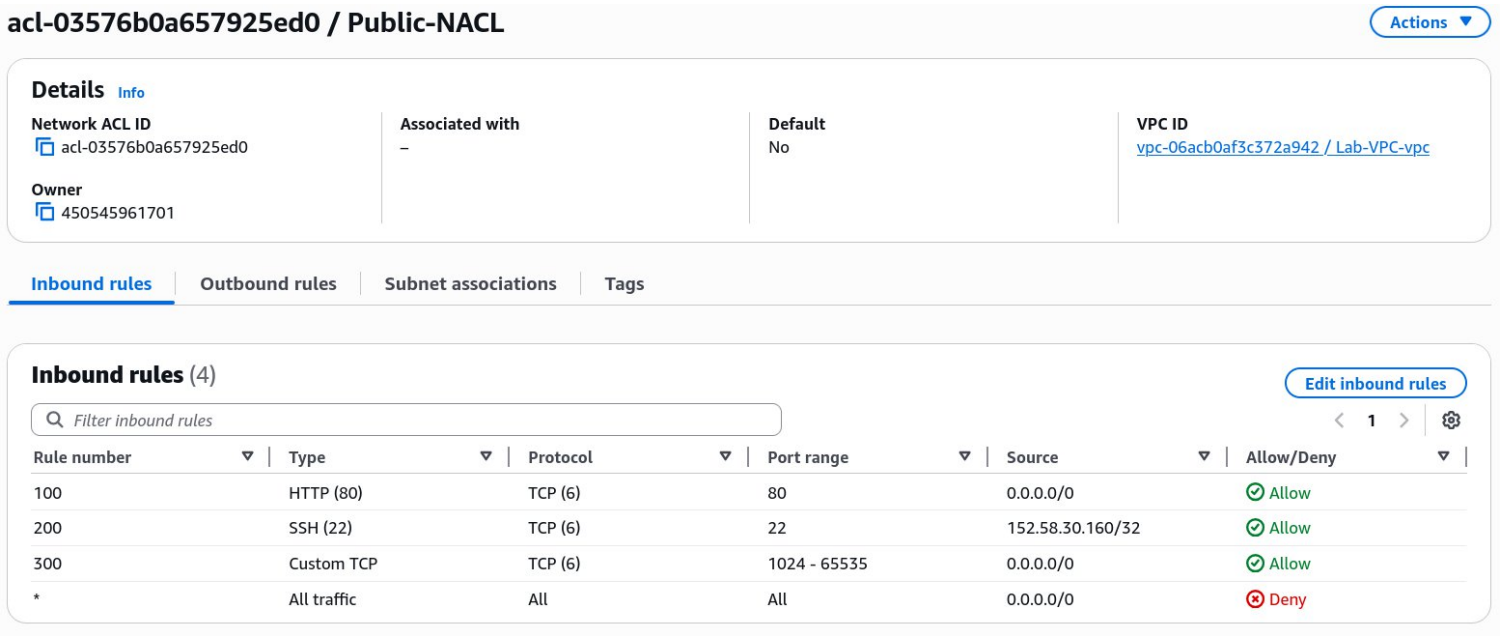
3. DB-SG (for Private Subnet):
   - Inbound Rules:
     - Type: MySQL/Aurora (3306) | Source: Select Web-SG (This restricts DB access only to the web servers).



B. Network ACLs (Subnet-Level Security)

NACLs are stateless; you must explicitly allow both inbound and outbound traffic.

1. Go to Network ACLs > Create network ACL.
2. Name it Public-NACL and associate it with your Public Subnet.
3. Inbound Rules:
   - Rule 100 | Port 80 (HTTP) | Source 0.0.0.0/0 | Allow
   - Rule 200 | Port 22 (SSH) | Source Your IP | Allow
   - Rule 300 | Port 1024-65535 (Ephemeral Ports) | Source 0.0.0.0/0 | Allow (Required for response traffic).
4. Outbound Rules:
   - Rule 100 | All Traffic | Destination 0.0.0.0/0 | Allow



Step 4: Validation Task

To confirm the lab is successful, perform these checks:

1. Launch a t2.micro instance in the Public Subnet using Web-SG.
2. Attempt to SSH into it using its Public IP.
3. Ping an external site (e.g., google.com) to verify the IGW and NACL outbound rules are working.

```
◄ 1m17s ❯ ssh -i MyLabKey.pem ec2-user@44.203.229.185
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sun Mar 29 11:07:32 2026 from 152.58.30.160
[ec2-user@ip-10-0-3-136 ~]$ ping -c 3 archlinux.org
PING archlinux.org (209.126.35.79) 56(84) bytes of data.
64 bytes from 209.126.35.79 (209.126.35.79): icmp_seq=1 ttl=51 time=6.67 ms
64 bytes from 209.126.35.79 (209.126.35.79): icmp_seq=2 ttl=51 time=6.56 ms
64 bytes from 209.126.35.79 (209.126.35.79): icmp_seq=3 ttl=51 time=6.58 ms

--- archlinux.org ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 6.560/6.603/6.665/0.044 ms
[ec2-user@ip-10-0-3-136 ~]$ |
```

Cleanup

Important: NAT Gateways and Elastic IPs incur hourly costs even if not used.

1. Delete any EC2 instances launched.
2. Go to Your VPCs, select Lab-VPC, and click Actions > Delete VPC. This will attempt to delete all associated subnets, gateways, and route tables automatically.